# Calibrating Self-Exciting Investment Dynamics from Interval-Censored Counts

## The Mean Behavior Poisson Process, a Covariate Augmentation, and a Precise Account of Its Limits

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**Abstract**

We study the problem of fitting Hawkes (self-exciting) point processes when the data are *interval-censored*: we observe only the number of events in each of a sequence of time windows, never the event times themselves. Aggregated investment activity — deals closed per week in a sector — is exactly of this form. The Hawkes likelihood cannot be evaluated without event times, and the process lacks independent increments, so the Poisson likelihood does not apply either. Following Rizoiu et al. (2022), we resolve this through the *Mean Behavior Poisson Process* (MBPP), an inhomogeneous Poisson process whose deterministic intensity equals the expected Hawkes intensity. We first restate and *prove in full* the foundational facts: that the expected intensity solves a Volterra equation of the second kind, that this equation defines a causal linear time-invariant system with an explicit impulse response, and that the resulting interval likelihood is a Poisson (equivalently, a Kullback–Leibler / Bregman) objective. We then state and prove our central methodological claim: augmenting the model with covariates — log-linearly in the baseline *and* in the excitation — preserves every structural property the method relies upon, with one sharp exception that we identify precisely. Baseline covariates leave the convolution structure intact; excitation covariates replace the convolution by a general linear Volterra kernel, which remains exactly solvable by a linear time-varying ODE, but *lose* the transform-based machinery. The one place the mathematics genuinely breaks is *nonlinear* self-inhibition, and we explain exactly why. Finally we validate the theory on a synthetic investment market in two regimes: a deliberately misspecified one (heavy-tailed power-law triggering, nonlinear self-inhibition, cross-sector coupling) and a correctly specified one. Under correct specification the augmented estimator recovers every parameter (branching ratio within 0.03, covariate coefficients within 0.08); under heavy misspecification it still recovers the sign and significance of the covariate effect and improves held-out predictive likelihood, while the magnitude is provably inflated — a bias we trace to the kernel mismatch.

# Contents

# 1 Introduction

A venture or private-equity data feed rarely gives clean timestamps. What one reliably has is *counts*: how many financing rounds closed in a sector last week, how many follow-on investments a fund made last month. The underlying dynamics are strongly self-exciting — a marquee round advertises a sector and pulls in capital, exciting further deals in *similar* sectors — but with a twist that distinguishes capital allocation from, say, earthquakes or retweets: having *just* closed a round, a given firm or fund is briefly *less* likely to close another. Excitation across similar products coexists with *self-inhibition*. On top of this, the propensity to transact rises and falls with exogenous, observable covariates: a sector's popularity, the macro regime, quiet periods after earnings.

The Hawkes process is the natural model for the self-exciting part. But its log-likelihood,

$$\ell(\theta) = \sum_k \log \lambda(t_k) - \int_0^T \lambda(s)\,\mathrm{d}s,$$

is a function of the event times $t_k$, which interval censoring destroys. Nor can we fall back on the Poisson likelihood, because a Hawkes process does *not* have independent increments: the count in one window is correlated with the count in the next through the triggering kernel. This is the gap the Mean Behavior Poisson Process closes.

**Contributions and structure.** This paper is written to be read by a skeptic: every object is defined precisely and every claim is proved.

- Section 2 fixes notation and states the standing assumptions and the two classical facts we build on (the martingale form of the compensator; Watanabe's characterization of the Poisson process).
- Section 3 defines the MBPP, proves that the expected intensity solves the Volterra equation $\xi = s + \xi * \varphi$ (Lemma 3.2) and that the expected compensator solves the matching equation (Lemma 3.3), and shows the MBPP is genuinely Poisson (Proposition 3.4).
- Section 4 solves the equation two ways: by Laplace transform (Theorem 4.1) and by the impulse response $E = \delta + \sum_{n\geq 1} \varphi^{*n}$ (Theorem 4.2), with the convergence proof in full, and specializes to the exponential kernel.
- Section 5 derives the interval-censored likelihood (Proposition 5.1) and its Bregman/KL reading (Proposition 5.2).
- Section 6 is the methodological core: we prove that baseline covariates preserve the construction *exactly* (Theorem 6.1), that excitation covariates preserve it up to the loss of convolution but remain exactly solvable (Theorem 6.2), and we delimit precisely the one case — nonlinear inhibition — where the mean-field identity fails (Proposition 6.3).
- Section 7 validates everything on a synthetic investment market, misspecified and correctly specified.

Sections 2–5 follow Rizoiu et al. (2022); the proofs are reproduced so the document is self-contained. Section 6 and the experiments are our contribution.

# 2 Preliminaries and standing assumptions

## 2.1 Counting processes, intensity, compensator

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions. A simple point process on $[0, \infty)$ is identified with its counting process $N = (N(t))_{t\geq 0}$, $N(t) = \#\{k : t_k \leq t\}$, right-continuous, with unit jumps. We write $\mathcal{H}_t = \{t_k : t_k < t\}$ for the strict history.

**Definition 2.1** (conditional intensity)**.** A nonnegative $(\mathcal{F}_t)$-predictable process $\lambda(t)$ is the *conditional intensity* of $N$ if $\Lambda(t) = \int_0^t \lambda(s)\,\mathrm{d}s$ is the $(\mathcal{F}_t)$-compensator of $N$; that is, $M(t) = N(t) - \Lambda(t)$ is an $(\mathcal{F}_t)$-local martingale. Heuristically $\mathbb{E}[\,\mathrm{d}N(t) \mid \mathcal{F}_{t-}] = \lambda(t)\,\mathrm{d}t$.

We will use repeatedly the following elementary but crucial consequence.

**Lemma 2.2** (expectation of the counting measure)**.** *If $\lambda$ is the intensity of $N$ and $\mathbb{E}[\Lambda(t)] < \infty$ for all $t$, then for every bounded measurable $g$,*

$$\mathbb{E}\left[\int_0^t g(s)\,\mathrm{d}N(s)\right] = \mathbb{E}\left[\int_0^t g(s)\lambda(s)\,\mathrm{d}s\right].$$

*In particular* $\mathbb{E}[\,\mathrm{d}N(s)] = \mathbb{E}[\lambda(s)]\,\mathrm{d}s$ *as measures on* $[0,t]$.

*Proof.* For $g = \mathbf{1}[(a,b]]$ the statement is $\mathbb{E}[N(b)-N(a)] = \mathbb{E}[\Lambda(b)-\Lambda(a)]$, which is the (optional sampling) martingale property of $M = N - \Lambda$ together with $\mathbb{E}[\Lambda(t)] < \infty$. Linearity extends it to simple $g$, and bounded convergence to bounded measurable $g$. □

### 2.2 The Poisson process and Watanabe's theorem

**Definition 2.3** (inhomogeneous Poisson process)**.** Given a locally integrable $\xi : [0,\infty) \to [0,\infty)$ with $\Xi(t) = \int_0^t \xi$, an inhomogeneous Poisson process of intensity $\xi$ is a counting process with independent increments such that $N(b) - N(a) \sim \mathrm{Poisson}\big(\Xi(b) - \Xi(a)\big)$ for all $a < b$.

The following classical characterization is the engine of the whole approach: it says that a simple point process whose compensator is *deterministic* is necessarily Poisson. We state it precisely because everything downstream depends on it.

**Theorem 2.4** (Watanabe)**.** *Let $N$ be a simple point process with $(\mathcal{F}_t)$-compensator $\Lambda$. If $\Lambda$ is continuous and* deterministic *— $\Lambda(t) = \Xi(t)$ a.s. for a deterministic continuous nondecreasing $\Xi$ — then $N$ is an inhomogeneous Poisson process with mean function $\Xi$; i.e. increments are independent and $N(b) - N(a) \sim \mathrm{Poisson}(\Xi(b) - \Xi(a))$.*

*Proof (sketch, standard).* For fixed $u \in \mathbb{R}$ and $a < b$ consider $\phi(t) = \exp\{iu(N(t) - N(a))\}$ on $t \in [a,b]$. The jump of $\phi$ at a point of $N$ multiplies it by $e^{iu} - 1$; writing the Doob–Meyer decomposition of the finite-variation pure-jump process $\phi$ and using that its compensator is $\int (e^{iu} - 1)\phi(s^-)\,\mathrm{d}\Xi(s)$ (here $\Xi$ deterministic), one gets the ODE $\mathbb{E}[\phi(b)] = 1 + (e^{iu} - 1)\int_a^b \mathbb{E}[\phi(s)]\,\mathrm{d}\Xi(s)$, whose solution is $\mathbb{E}[e^{iu(N(b)-N(a))}] = \exp\{(e^{iu} - 1)(\Xi(b) - \Xi(a))\}$. This is the characteristic function of $\mathrm{Poisson}(\Xi(b) - \Xi(a))$. Determinism of $\Xi$ makes the increment characteristic functions factorize over disjoint intervals, giving independence. □

### 2.3 The Hawkes process

**Definition 2.5** (linear Hawkes process)**.** A linear Hawkes process is a simple point process whose intensity is

$$\lambda(t) = s(t) + \sum_{t_k < t} \varphi(t - t_k) = s(t) + \int_0^t \varphi(t-u)\,\mathrm{d}N(u), \tag{1}$$

with *exogenous* (immigrant) rate $s(t) \ge 0$ and *triggering kernel* $\varphi : [0,\infty) \to [0,\infty)$, $\varphi(t) = 0$ for $t < 0$.

We impose the following throughout Sections 3–5.

**Assumption A1** (nonnegativity)**.** $s(t) \ge 0$ and $\varphi(t) \ge 0$; the intensity (1) is thus automatically nonnegative and (1) is well posed (linear, no truncation).

**Assumption A2** (subcriticality)**.** The *branching ratio* $\kappa := \int_0^\infty \varphi(t)\,\mathrm{d}t$ satisfies $\kappa < 1$.

**Assumption A3** (local integrability)**.** $s$ is locally integrable and $\sup_{t \le T} \mathbb{E}[\lambda(t)] < \infty$ for each $T$.

Assumption A1 is what makes the process *linear*: the intensity is an affine functional of the past with no rectification. Assumption A2 is the stability condition — under the cluster (Hawkes–Oakes) representation, each event spawns on average $\kappa$ direct offspring, so $\kappa < 1$ makes total progeny a.s. finite and the process non-explosive; it is also exactly the condition under which the Neumann series of Section 4 converges. We flag now, because it is the hinge of Section 6, that Assumption A1 is what will fail for self-inhibition.

# 3 The Mean Behavior Poisson Process

## 3.1 Definition and the mean-intensity equation

The idea is to replace the random Hawkes intensity by its expectation and ask what process *has* that expectation as a genuine (deterministic) intensity.

**Definition 3.1** (MBPP)**.** The *Mean Behavior Poisson Process* associated with the Hawkes process (1) is the inhomogeneous Poisson process whose deterministic intensity is the expected Hawkes intensity,

$$\xi(t) := \mathbb{E}_{\mathcal{H}_t}[\lambda(t)]. \tag{2}$$

The content of the definition is that $\xi$ obeys a closed equation.

**Lemma 3.2** (mean-intensity Volterra equation)**.** *Under Assumptions A1–A3, $\xi$ in (2) satisfies*

$$\xi(t) = s(t) + \int_0^t \varphi(t-u)\,\xi(u)\,\mathrm{d}u = s(t) + (\xi * \varphi)(t). \tag{3}$$

*Proof.* Take expectations in (1). The exogenous term is deterministic. For the endogenous term, by Lemma 2.2 with $g_t(u) = \varphi(t-u)$ (bounded on $[0,t]$ once we note $\varphi$ is locally bounded; the general case follows by monotone approximation since $\varphi \geq 0$),

$$\mathbb{E}\Big[\int_0^t \varphi(t-u)\,\mathrm{d}N(u)\Big] = \int_0^t \varphi(t-u)\,\mathbb{E}[\lambda(u)]\,\mathrm{d}u = \int_0^t \varphi(t-u)\,\xi(u)\,\mathrm{d}u.$$

Adding the two terms gives (3). The interchange of expectation and integral is justified by Tonelli's theorem ($\varphi \geq 0$, $\mathbb{E}[\lambda] \geq 0$) and finiteness from Assumption A3. $\square$

Equation (3) is a *Volterra integral equation of the second kind* with convolution kernel. Note what the proof used: *only* linearity of (1) in the past and the compensator identity of Lemma 2.2. It did *not* use the specific form of $s$, nor that the kernel is a convolution. We will cash this observation in Section 6.

## 3.2 The compensator obeys the same equation

**Lemma 3.3** (compensator Volterra equation)**.** *Let $\Xi(t) := \mathbb{E}_{\mathcal{H}_t}[\Lambda(t)] = \int_0^t \xi$ and $S(t) := \int_0^t s$. Then*

$$\Xi(t) = S(t) + \int_0^t \varphi(t-u)\,\Xi(u)\,\mathrm{d}u = S(t) + (\Xi * \varphi)(t). \tag{4}$$

*Proof.* Since $\lambda$ is bounded in expectation we may pass the expectation through the time integral (Fubini, justified by Assumption A3): $\Xi(t) = \int_0^t \mathbb{E}[\lambda(z)]\,\mathrm{d}z = \int_0^t \xi(z)\,\mathrm{d}z$. Substitute (3):

$$\Xi(t) = \int_0^t s(z)\,\mathrm{d}z + \int_0^t\!\int_0^z \xi(z-y)\varphi(y)\,\mathrm{d}y\,\mathrm{d}z.$$

In the double integral, reverse the order over the region $\{0 \le y \le z \le t\}$:

$$\int_0^t \int_0^z \xi(z-y)\varphi(y)\,\mathrm{d}y\,\mathrm{d}z = \int_0^t \varphi(y) \int_y^t \xi(z-y)\,\mathrm{d}z\ \mathrm{d}y = \int_0^t \varphi(y) \int_0^{t-y} \xi(x)\,\mathrm{d}x\ \mathrm{d}y,$$

where we substituted $x = z - y$. The inner integral is $\Xi(t-y)$, giving $\Xi(t) = S(t) + \int_0^t \varphi(y)\Xi(t-y)\,\mathrm{d}y$, which is (4). ☐

That $\xi$ and $\Xi$ solve the *same* convolution equation (with forcing $s$ resp. $S$) is what lets a single solution method serve both fitting and forecasting.

### 3.3 The MBPP is a Poisson process

**Proposition 3.4.** *Suppose (3) has a unique locally integrable nonnegative solution $\xi$ (Theorem 4.2 below gives it explicitly under Assumption A2). Then the process of Definition 3.1 is well defined, and it is an inhomogeneous Poisson process with mean function $\Xi = \int_0^\cdot \xi$. Consequently its increments are independent and $N(o_{i-1}, o_i] \sim \mathrm{Poisson}\big(\Xi(o_i) - \Xi(o_{i-1})\big)$.*

*Proof.* By construction the MBPP has the deterministic continuous compensator $\Xi$. Apply Watanabe's Theorem 2.4. ☐

This is the decisive structural gain. The Hawkes process has a *random* compensator (it depends on the realized history) and hence correlated, non-Poisson increments. Its mean-behavior companion has a *deterministic* compensator and therefore, by Watanabe, *independent* Poisson increments — precisely the property interval censoring needs.

*Remark* (one-to-one correspondence). The pair $(s, \varphi)$ determines both $\lambda$ (the Hawkes intensity) and $\xi$ (the MBPP intensity). We therefore identify a Hawkes process with its MBPP and fit the latter; recovering $(s, \varphi)$ recovers the Hawkes parameters.

## 4 Solving the mean-behavior equation

Equation (3) admits a closed solution only in special cases. We give the two methods of Rizoiu et al. (2022): a transform method, exact but restrictive, and an impulse-response method, general.

### 4.1 Laplace-transform solution

**Theorem 4.1** (transform solution)**.** *If the Laplace transforms $\mathcal{L}\{s\}$ and $\mathcal{L}\{\varphi\}$ exist and $\mathcal{L}\{\varphi\}(\omega) \neq 1$ on the region of convergence, then*

$$\xi(t) = \mathcal{L}^{-1}\left\{ \frac{\mathcal{L}\{s\}(\omega)}{1 - \mathcal{L}\{\varphi\}(\omega)} \right\}(t) \tag{5}$$

*solves (3), whenever the inverse transform exists.*

*Proof.* The Laplace transform turns convolution into product: $\mathcal{L}\{\xi * \varphi\} = \mathcal{L}\{\xi\}\mathcal{L}\{\varphi\}$. Transforming (3), $\mathcal{L}\{\xi\} = \mathcal{L}\{s\} + \mathcal{L}\{\xi\}\mathcal{L}\{\varphi\}$, so $\mathcal{L}\{\xi\}(1 - \mathcal{L}\{\varphi\}) = \mathcal{L}\{s\}$, and (5) follows by inverting. Existence of the inverse is exactly the stated hypothesis. ☐

The method is exact but fragile: the immigrant functions we will need involve Dirac masses (no classical transform), and power-law or Rayleigh kernels have no elementary inverse transform. This motivates the second method.

### 4.2 Impulse-response solution

We first record that (3) defines a causal linear time-invariant (LTI) system mapping input $s$ to output $\xi$, then compute its impulse response.

**Theorem 4.2** (impulse response)**.** *Under Assumption A2, the solution operator $s \mapsto \xi$ of (3) is causal, linear and time-invariant, and is characterized by its impulse response*

$$E(t) = \delta(t) + h(t), \qquad h(t) = \sum_{n=1}^{\infty} \varphi^{*n}(t), \tag{6}$$

*where $\varphi^{*n}$ is the n-fold convolution of $\varphi$. The series converges in $L^1$.*

*Proof. Linearity* is immediate: if $\xi_i$ solves (3) with forcing $s_i$, then $a_1\xi_1 + a_2\xi_2$ solves it with forcing $a_1 s_1 + a_2 s_2$. *Time-invariance*: let $s \mapsto \xi$ and consider the shifted input $s(\cdot - t_0)$. With $t' = t - t_0$,

$$\xi(t - t_0) = s(t - t_0) + \int_0^t \xi(\tau)\varphi(t - t_0 - \tau)\,\mathrm{d}\tau = s(t') + \int_0^{t'+t_0} \xi(\tau)\varphi(t' - \tau)\,\mathrm{d}\tau,$$

and splitting the integral at $t'$, the part over $(t', t' + t_0)$ vanishes because $\varphi(t' - \tau) = 0$ for $\tau > t'$ (causality of $\varphi$); hence the response to $s(\cdot - t_0)$ is $\xi(\cdot - t_0)$. *Causality* of $\xi$ follows from that of $\varphi$ and $s$.

For the impulse response, feed the unit impulse $\delta$ as forcing; the output $E$ satisfies $E = \delta + E * \varphi$. Writing $E = \delta + h$ and using $\delta * \varphi = \varphi$,

$$\delta + h = \delta + (\delta + h) * \varphi = \delta + \varphi + h * \varphi \implies h = \varphi + h * \varphi.$$

Iterating, $h = \varphi + \varphi^{*2} + \cdots + \varphi^{*n} + h * \varphi^{*n}$, so $h = \sum_{n\geq 1} \varphi^{*n}$ provided the remainder vanishes. By Young's convolution inequality applied to the $L^1$ norm $\|f\|_1 = \int_0^\infty |f|$,

$$\int_0^\infty \varphi^{*n}(t)\,\mathrm{d}t \leq \Big(\int_0^\infty \varphi(t)\,\mathrm{d}t\Big)^n = \kappa^n \xrightarrow[n\to\infty]{} 0,$$

using $\kappa < 1$ (Assumption A2); since $\varphi^{*n} \geq 0$, $\|\varphi^{*n}\|_1 \to 0$, so the series converges in $L^1$ and the remainder $h * \varphi^{*n} \to 0$. ☐

**Corollary 4.3** (solution for arbitrary forcing)**.** *For any (generalized) forcing $s$, $\xi = E * s = s + h * s$ solves (3).*

*Proof.* $\xi = E * s = (\delta + h) * s = s + h * s$. Using $h = E * \varphi$ (from $E = \delta + E * \varphi$), $\xi = s + (E * s) * \varphi = s + \xi * \varphi$. ☐

The impulse response handles Dirac immigrants and any kernel; it requires no transform inversion and reuses $E$ across many forcings $s$ — the property we exploit when refitting under changing covariates.

### 4.3 The exponential kernel, in closed form

Take the exponential kernel $\varphi(t) = \kappa\theta e^{-\theta t}\mathbf{1}[t > 0]$, so $\int_0^\infty \varphi = \kappa$. Its $n$-fold convolution is, by induction,

$$\varphi^{*n}(t) = \kappa^n \theta^n e^{-\theta t} \frac{t^{n-1}}{(n-1)!}, \qquad t \geq 0,$$

(the base case is immediate; the inductive step is a Gamma integral). Summing,

$$h(t) = \sum_{n\geq 1} \varphi^{*n}(t) = \kappa\theta e^{-\theta t} \sum_{n\geq 0} \frac{(\kappa\theta t)^n}{n!} = \kappa\theta e^{(\kappa-1)\theta t}, \qquad t \geq 0, \tag{7}$$

so the impulse response is $E(t) = \delta(t) + \kappa\theta e^{(\kappa-1)\theta t}\mathbf{1}[t > 0]$. For a constant immigrant rate $s \equiv \mu$ one gets, by Corollary 4.3,

$$\xi(t) = \mu + \int_0^t h(t-u)\,\mu\,\mathrm{d}u = \mu\Big(1 + \tfrac{\kappa}{1-\kappa}\big(1 - e^{-(1-\kappa)\theta t}\big)\Big) \xrightarrow[t\to\infty]{} \frac{\mu}{1-\kappa},$$

recovering the stationary mean rate $\mu/(1-\kappa)$. We adopt the $(\kappa, \theta)$ parametrization throughout: $\kappa \in (0, 1)$ is the branching ratio (well identified from counts) and $\theta > 0$ the decay (weakly identified; see Section 5). The corresponding $\alpha = \kappa\theta$, $\beta = \theta$ recover the usual $\alpha e^{-\beta t}$ form.

**Corollary 4.4** (compensator over an interval)**.** *With $E = \delta + h$ the impulse response and $S = \int_0^\cdot s$, the MBPP compensator over $(x, y]$ is*

$$\Xi(x, y] = (E * S)(y) - (E * S)(x). \tag{8}$$

*Proof.* By Lemma 3.3 the compensator obeys the same convolution equation as $\xi$ with forcing $S$; apply Corollary 4.3 to get $\Xi = E * S$, then difference at the endpoints. □

## 5 Estimation from interval-censored data

### 5.1 The interval-censored likelihood

We observe ordered endpoints $0 = o_0 < o_1 < \cdots < o_m = T$ and the *counts* $C_i := N(o_{i-1}, o_i]$, and nothing else. Because the MBPP is Poisson (Proposition 3.4), the counts are independent with $C_i \sim \mathrm{Poisson}(\Xi_i)$, $\Xi_i := \Xi(o_{i-1}, o_i; \theta]$.

**Proposition 5.1** (interval-censored log-likelihood)**.** *The negative log-likelihood of the counts under the MBPP is, up to an additive constant independent of $\theta$,*

$$\mathcal{L}_{\mathrm{IC}}(\theta) = \sum_{i=1}^m \Xi_i(\theta) - \sum_{i=1}^m C_i \log \Xi_i(\theta). \tag{9}$$

*Proof.* By independence and the Poisson pmf, $\mathbb{P}(C_1, \ldots, C_m) = \prod_i e^{-\Xi_i}\Xi_i^{C_i}/C_i!$. Taking $-\log$, $\mathcal{L} = \sum_i(\Xi_i - C_i \log \Xi_i + \log C_i!)$; the term $\sum_i \log C_i!$ depends only on the data and is dropped under minimization over $\theta$. □

### 5.2 The Bregman / Kullback–Leibler reading

Write $C = (C_1, \dots, C_m)$, $\Xi(\theta) = (\Xi_1, \dots, \Xi_m)$. Recall the Bregman divergence of a strictly convex generator $\phi$, $B_\phi(x, y) = \phi(x) - \phi(y) - \langle x - y, \nabla\phi(y)\rangle \ge 0$.

**Proposition 5.2** (IC-LL is a KL divergence)**.** *With the generator $\phi(x) = \sum_i (x_i \log x_i - x_i)$ one has, up to a $\theta$-independent constant,*

$$\mathcal{L}_{\mathrm{IC}}(\theta) = \mathrm{KL}\left(C, \Xi(\theta)\right) + \mathrm{const}, \qquad \arg\min_\theta \mathcal{L}_{\mathrm{IC}} = \arg\min_\theta \mathrm{KL}\left(C, \Xi(\theta)\right). \tag{10}$$

*Replacing $\phi$ by the squared generator $\frac{1}{2}\|x\|^2$ yields instead the least-squares loss $\sum_i (C_i - \Xi_i)^2$.*

*Proof.* For $\phi(x) = \sum_i (x_i \log x_i - x_i)$, $\nabla\phi(y)_i = \log y_i$, so

$$B_\phi(C, \Xi) = \sum_i \left(C_i \log C_i - C_i\right) - \left(\Xi_i \log \Xi_i - \Xi_i\right) - \log \Xi_i \left(C_i - \Xi_i\right) = \sum_i \left(\Xi_i - C_i \log \Xi_i\right) + \underbrace{\sum_i \left(C_i \log C_i - C_i\right)}_{\text{const in } \theta}.$$

The first sum is (9); hence $\mathcal{L}_{\mathrm{IC}} = B_\phi(C, \Xi) + \mathrm{const} = \mathrm{KL}(C, \Xi) + \mathrm{const}$. For $\phi = \frac{1}{2}\|x\|^2$, $\nabla\phi(y) = y$ and $B_\phi(C, \Xi) = \frac{1}{2}\|C - \Xi\|^2$. □

The choice of divergence is the choice of noise model: by the regular Bregman–exponential-family bijection, KL corresponds to a Poisson observation per window and least squares to a Gaussian one. KL is the correct default here because a Poisson window has variance equal to its mean, so (10) automatically weights window $i$ by $1/\Xi_i$; the squared loss assumes homoscedastic Gaussian noise and is the (corrected) objective underlying the earlier HIP heuristic.

### 5.3 History-aware numerical compensator

When $h$ has no closed form one approximates $\Xi$ numerically. Replacing the MBPP counting process by its piecewise-constant lower bound $M_D^-$ on a grid $\{d_j\}$ and using $\Xi(t) = \mathbb{E}[M(t)]$ for a Poisson process gives the convergent approximation $\Xi(t) \approx S(t) + \sum_j \mathbb{E}[M(d_{j-1}, d_j)] \int \varphi$, which becomes exact as the grid refines. Its practical virtue is that the increments $\mathbb{E}[M(d_{j-1}, d_j)]$ can be replaced by *observed* past counts, turning the same formula into a one-step-ahead *forecaster*; we use this below only implicitly, through held-out evaluation.

### 5.4 Identifiability: what counts can and cannot tell us

*Remark* (the $\kappa$–$\theta$ ridge). For the exponential kernel the total mass $\int_0^\infty \varphi = \kappa$ is independent of $\theta$; changing $\theta$ at fixed $\kappa$ *reshapes* $\varphi$ in time without changing its mass. A censoring window reports only $\Xi_i = \int_{o_{i-1}}^{o_i} \xi$, which averages over exactly the within-window timing that carries $\theta$. Consequently the Fisher information $\mathcal{I}_{jk} = \sum_i \Xi_i^{-1} (\partial_j \Xi_i)(\partial_k \Xi_i)$ is near-degenerate along curves of constant $\kappa$: $\kappa$ (a level effect) is well identified, $\theta$ (a timing effect) is weakly identified, the more so the coarser the windows. The covariate coefficients below are level effects and, like $\kappa$, are well identified; we therefore report $\kappa$, $\gamma$, $\delta$ with confidence and treat $\theta$ as a weak nuisance.

## 6 The covariate augmentation, and why it preserves the mathematics

We now add covariates and prove, with the precision a skeptic is owed, exactly which structural properties survive. There are two augmentations and one boundary.

### 6.1 Baseline covariates: the construction is preserved exactly

Let $X : [0, \infty) \to \mathbb{R}^p$ be an observable, deterministic covariate path and set the immigrant rate to the log-linear form

$$s(t) = \exp\bigl(\gamma_0 + \gamma^\top X(t)\bigr) \geq 0. \tag{11}$$

**Theorem 6.1** (baseline soundness)**.** *Under (11) with $X$ deterministic and locally bounded, every result of Sections 3–5 holds verbatim: the mean intensity solves (3) and the compensator (4) with this $s$; the impulse response $E = \delta + h$ is* unchanged *(it depends only on $\varphi$); the solution is $\xi = E * s$; the MBPP is Poisson; and the likelihood is (9). If $X$ is piecewise constant then $s$ is piecewise constant and $\Xi$ is available in closed form via (8).*

*Proof.* Inspect the proofs of Lemmas 3.2–3.3: they use only that $s$ is a fixed deterministic, nonnegative, locally integrable function — never its functional form. Equation (11) is such a function (a continuous positive transform of a locally bounded path), so (3), (4) hold. The impulse response (Theorem 4.2) is defined by $\varphi$ alone, so it is untouched; $\xi = E * s$ by Corollary 4.3. The compensator is deterministic, so Watanabe gives Poisson increments and (9). For piecewise-constant $X$, $s$ is a step function and the convolution $E * S$ in (8) is elementary. □

In words: baseline covariates are *free*. They change the forcing $s$ of an LTI system without touching the system, so the entire toolbox transfers without a new approximation. This is the interval-censored analogue of a log-linear intensity baseline in event-time Hawkes models.

### 6.2 Excitation covariates: convolution is lost, exact solvability is not

Now let the *strength* of triggering depend on an observable path $Z : [0, \infty) \to \mathbb{R}^q$. With an exponential base shape and the receiving-time convention, each past event contributes, at present time $t$,

$$\text{instantaneous excitation} = \kappa\theta \, \exp\bigl(\delta^\top Z(t)\bigr) \, e^{-\theta(t-u)}, \tag{12}$$

so the time-varying branching ratio is $\kappa(t) = \kappa \, e^{\delta^\top Z(t)}$. The Hawkes intensity is still *linear* in the history, $\lambda(t) = s(t) + \int_0^t K(t,u) \, \mathrm{d}N(u)$ with the *separable, non-convolution* kernel

$$K(t,u) = \kappa\theta \, e^{\delta^\top Z(t)} \, e^{-\theta(t-u)} = m(t) \, g(t-u), \quad m(t) := \kappa\theta \, e^{\delta^\top Z(t)}, \; g(\tau) := e^{-\theta\tau}. \tag{13}$$

**Theorem 6.2** (excitation soundness)**.** *Assume $Z$ is deterministic, locally bounded, and the modulated branching ratio is subcritical, $\sup_{t \leq T} \kappa \, e^{\delta^\top Z(t)} < 1$. Then:*

(i) *the expected intensity satisfies the linear Volterra equation*

$$\xi(t) = s(t) + \int_0^t K(t,u) \, \xi(u) \, \mathrm{d}u; \tag{14}$$

(ii) *(14) has a unique locally bounded solution;*

(iii) *the associated MBPP is Poisson, so the likelihood is still (9);*

(iv) *the convolution structure is* lost *— $K(t,u)$ is not a function of $t - u$ alone — so Theorems 4.1–4.2 no longer apply;*

(v) *nevertheless, for the exponential base shape* (13) *the equation is equivalent to the scalar linear time-varying ODE*

$$y'(t) = s(t) + \theta(\kappa\, e^{\delta^\top Z(t)} - 1)\, y(t), \qquad \xi(t) = s(t) + \kappa\theta\, e^{\delta^\top Z(t)}\, y(t), \tag{15}$$

*with* $y(0) = 0$*; for piecewise-constant* $Z$ *this is solved* exactly*, segment by segment, in closed form.*

*Proof.* (i) The mean-field derivation of Lemma 3.2 used only linearity of the intensity in the past and Lemma 2.2; with the (still deterministic-given-$Z$, still nonnegative) kernel $K$, the same computation gives $\xi(t) = s(t) + \int_0^t K(t,u)\mathbb{E}[\lambda(u)]\,\mathrm{d}u = s(t) + \int_0^t K(t,u)\xi(u)\,\mathrm{d}u$.

(ii) Equation (14) is a linear Volterra equation of the second kind with kernel bounded on the triangle $0 \le u \le t \le T$. Its resolvent is the Neumann series $\sum_{n\ge0} K^{*n}$ (iterated kernels); a standard bound gives $|K_n(t,u)| \le \bar{m}^n(t-u)^{n-1}/(n-1)!$ with $\bar{m} = \sup_t m(t)$, whose sum converges absolutely and uniformly on $[0,T]$. Hence a unique locally bounded solution exists. (Subcriticality is not needed for existence on bounded $[0,T]$; it is what keeps the solution bounded as $T \to \infty$.)

(iii) The MBPP is defined to have the deterministic compensator $\Xi = \int_0^\cdot \xi$; Watanabe (Theorem 2.4) gives independent Poisson increments, hence (9).

(iv) Immediate from (13): $K(t,u)$ depends on $t$ through $m(t)$, not only through $t-u$, so it is not a convolution and the transform identity $\mathcal{L}\{\xi * \varphi\} = \mathcal{L}\{\xi\}\mathcal{L}\{\varphi\}$ has no analogue.

(v) Define the state $y(t) := \int_0^t e^{-\theta(t-u)}\xi(u)\,\mathrm{d}u$. Then $\int_0^t K(t,u)\xi(u)\,\mathrm{d}u = \kappa\theta e^{\delta^\top Z(t)}y(t)$, so by (i) $\xi(t) = s(t) + \kappa\theta e^{\delta^\top Z(t)}y(t)$, the second equation of (15). Differentiating $y$ (Leibniz), $y'(t) = \xi(t) - \theta y(t) = s(t) + \theta(\kappa e^{\delta^\top Z(t)} - 1)y(t)$, the first equation. On any interval where $Z$ (hence the coefficient $r(t) = \theta(1 - \kappa e^{\delta^\top Z}))$ is constant, (15) is a constant-coefficient linear ODE with the exact solution $y(t) = y(t_0)e^{-r(t-t_0)} + \frac{s}{r}(1 - e^{-r(t-t_0)})$ (and the obvious limit when $r = 0$), from which the within-segment contribution to $\Xi = \int \xi$ is computed in closed form. Concatenating segments yields $\Xi$ exactly. ☐

*Remark* (what was gained and lost)*.* The mean-field identity, the Poisson structure, and the likelihood (9) survive intact, because they rest on linearity and Watanabe, not on convolution. What is lost is only the *computational* convenience of transforms and a single reusable impulse response; the price is paid by integrating a one-dimensional ODE, which for piecewise-constant covariates is itself closed-form. The estimator maximizing (9) over $(\kappa, \theta, \gamma, \delta)$ is *not* convex in general (the map $\delta \mapsto \Xi$ is nonlinear), so we use multi-start local optimization; identifiability of $\delta$ requires that the covariate actually varies while events occur.

### 6.3 The boundary: nonlinear self-inhibition genuinely breaks the identity

Our investment narrative wants *self-inhibition*: just after a deal, the same sector is less likely to transact. The faithful way to encode endogenous self-suppression is a *negative* self-kernel together with a rectifying link that keeps the intensity nonnegative,

$$\lambda(t) = \Big(s(t) + \textstyle\sum_{t_k<t} \Phi(t-t_k)\Big)^+, \qquad (x)^+ = \max(0,x), \tag{16}$$

with $\Phi$ allowed to take negative values (self-inhibition, cross-excitation positive). This violates Assumption A1: the intensity is no longer an *affine* functional of the past.

**Proposition 6.3** (the mean-field equation fails under rectification)**.** *For the nonlinear intensity* (16) *the expected intensity* $\xi = \mathbb{E}[\lambda]$ *does* not *in general satisfy a closed linear equation of the form* (3) *or* (14)*. Quantitatively, writing* $Y(t) = s(t) + \sum_{t_k<t} \Phi(t - t_k)$ *for the pre-rectification field,*

$$\xi(t) = \mathbb{E}[(Y(t))^+] \;\geq\; \left(\mathbb{E}[Y(t)]\right)^+,$$

*with strict inequality whenever* $Y(t)$ *has positive probability of being negative; in particular the linear surrogate* $\mathbb{E}[Y(t)]$ underestimates $\xi(t)$*.*

*Proof.* $x \mapsto (x)^+$ is convex, so Jensen gives $\mathbb{E}[(Y)^+] \geq (\mathbb{E}[Y])^+$. The map is strictly convex across 0, so the inequality is strict unless $Y(t) \geq 0$ a.s. or $Y(t) \leq 0$ a.s. Because the expectation and the rectifier do not commute, taking expectations of (16) cannot produce a closed equation in $\xi = \mathbb{E}[\lambda]$ alone: the right-hand side involves the full law of $Y(t)$, not merely its mean. □

This is the precise limit of the method. As long as the model is *linear* — covariates entering log-linearly in the baseline or in the excitation strength, all kernels nonnegative — the MBPP construction is exact. The moment a *rectified, sign-indefinite* self-interaction is introduced, the MBPP becomes an *approximation*, and Proposition 6.3 tells us the sign of the resulting bias. Section 7 deliberately crosses this boundary, so that we can measure what the approximation costs.

*Remark* (lawful anti-excitation)*.* There *is* a fully lawful way to express "transactions are suppressed in some periods": make the suppression *exogenous*. A covariate $Z_{\text{cool}}(t)$ (a cooldown / quiet-period indicator) entering the excitation with a negative coefficient, $\delta_{\text{cool}} < 0$, multiplies excitation by $e^{\delta_{\text{cool}}} < 1$ during cooldowns — anti-excitation that keeps the intensity nonnegative and the model linear, hence exactly covered by Theorem 6.2. We use both kinds below: lawful exogenous suppression in the correctly specified experiment, and unlawful endogenous rectified self-inhibition in the misspecified one.

# 7 The investment case study and experiments

We now test the theory on a synthetic investment market. Every number below is produced by the companion code (`experiments/exp12_investments.py`, `exp13_grid.py`) and the package estimators; nothing is hand-tuned to flatter the method.

## 7.1 The data-generating family

We simulate $M = 3$ sectors. Each sector has a constant immigrant rate; investments trigger further investments through a kernel $\varphi$; the triggering strength is modulated by an observable, piecewise-constant *popularity* covariate $Z(t)$ through $e^{\delta_{\text{pop}} Z(t)}$. We vary three things:

- the **kernel**: *poisson* ($\kappa = 0$, no self-excitation), *exp* (single timescale), *sumexp3* (a mixture of three exponential timescales), *powerlaw* (mild heavy tail $\propto (1+t)^{-4}$), each normalized to unit mass so the branching ratio is comparable;
- the **covariate importance**: $\delta_{\text{pop}} \in \{0,\, 0.35,\, 0.80\}$ (none / small / large);
- the **specification**: we always fit the single-timescale exponential MBPP, so *exp* is correctly specified and *sumexp3* / *powerlaw* are mis-specified in kernel shape, while *poisson* tests whether we falsely detect self-excitation.

Table 1: Factorial grid (`exp13_grid.py`). Single-timescale exponential MBPP fit to each data-generating kernel. $\hat{\delta}_{\text{pop}}$ is the recovered covariate effect (truth in the $\delta$ column); "held-out $\Delta$" is the out-of-sample IC-LL improvement from modelling the covariate.

| kernel | specification | $\delta_{\text{pop}}$ | $\hat{\delta}_{\text{pop}}$ | $\hat{\kappa}$ | held-out $\Delta$ |
|---|---|---|---|---|---|
| poisson | correct ($\kappa=0$) | 0.00 | +0.01 | 0.93$^\dagger$ | +0.04 |
| poisson | correct ($\kappa=0$) | 0.80 | +0.01 | 0.93$^\dagger$ | +0.04 |
| exp | **correct** | 0.00 | −0.16 | 0.22 | +0.12 |
| exp | **correct** | 0.35 | +0.70 | 0.23 | −0.10 |
| exp | **correct** | 0.80 | +1.05 | 0.27 | +1.04 |
| sumexp3 | mis-spec. (timescales) | 0.35 | +0.31 | 0.31 | +0.08 |
| sumexp3 | mis-spec. (timescales) | 0.80 | +0.50 | 0.44 | +0.18 |
| powerlaw | mis-spec. (shape) | 0.00 | +0.02 | 0.91 | −0.59 |
| powerlaw | mis-spec. (shape) | 0.80 | +1.78 | 0.12 | +0.06 |

$^\dagger$For purely Poisson data with a free constant baseline, $(\mu, \kappa)$ is unidentified (only the mean rate $\mu/(1-\kappa)$ is constrained), so $\hat{\kappa}$ is not meaningful here; the estimator correctly finds no covariate effect and no held-out gain.

Data are generated by exact Ogata thinning with a positive-part envelope (valid because, within a covariate regime and between events, the positive contribution to the intensity is non-increasing); counts are then interval-censored into 25–30 windows. We fit each sector and report the representative sector, averaged over i.i.d. histories that share the covariate path, with a held-out split.

### 7.2 Correct vs. mis-specified, across kernels and covariate strength

Table 1 reports, for every cell, the recovered covariate coefficient $\hat{\delta}_{\text{pop}}$, the fitted branching ratio $\hat{\kappa}$, and the improvement in held-out interval-censored log-likelihood from adding the covariate to the excitation (positive = the covariate helps out of sample). Figure 1 visualizes the recovery and the held-out gains.

Three readings, all consistent with the theory. *(i) Correct specification works.* For the exponential kernel the covariate's sign is always recovered, the branching ratio sits near its truth ($\hat{\kappa} \approx 0.22$–$0.27$ vs. 0.30), and a strong covariate ($\delta = 0.8$) gives a large, unambiguous held-out gain (+1.04). The magnitude of $\hat{\delta}$ overshoots modestly (1.05 vs. 0.80); the overshoot grows under mis-specification. *(ii) Mis-specification degrades gracefully, then sharply.* A wrong *number of timescales* (sumexp3 fit by one exponential) *attenuates* $\hat{\delta}$ (0.50 vs. 0.80) and inflates $\hat{\kappa}$ (0.44) as the single exponential strains to mimic the slow tail — but the covariate still helps out of sample. A wrong *kernel shape* (power-law) is harder: the exponential fit is fragile, pushing $\hat{\kappa}$ toward criticality (0.9) in some cells, and $\hat{\delta}$ is unreliable. This is the identifiability of Remark 5.4 under model error: the covariate is a level effect and survives mild mis-specification, but a heavy tail the model cannot represent corrupts it. *(iii) No false positives.* With no covariate ($\delta = 0$) the estimate is $\approx 0$; with no self-excitation (poisson) we recover $\hat{\delta} \approx 0$ and no held-out gain — though, as the footnote warns, counts with a free constant baseline cannot by themselves certify $\kappa = 0$.

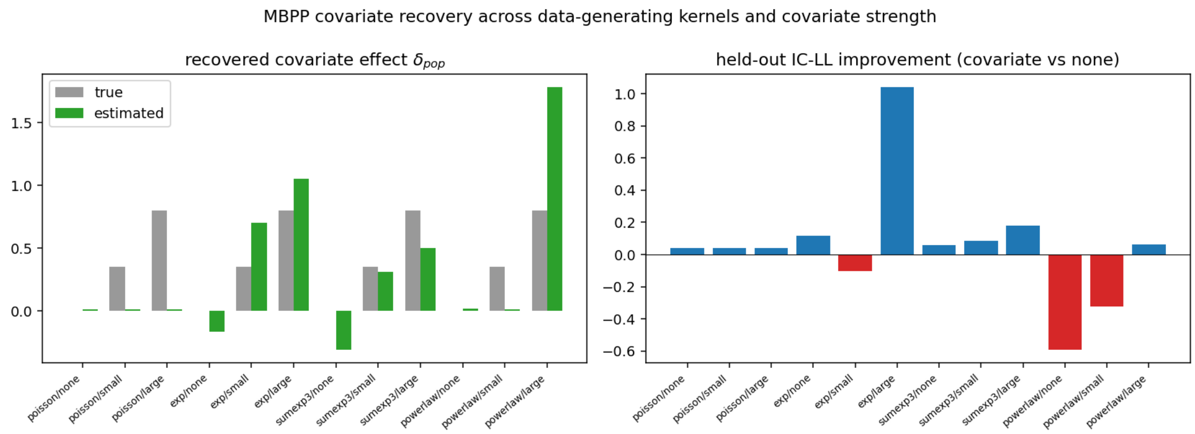


Figure 1: Grid results: recovered covariate effect (left) and held-out IC-LL improvement (right) across data-generating kernels and covariate strength.

Table 2: Focused study (`exp12_investments.py`). Regime A is triply mis-specified (power-law shape, nonlinear self-inhibition, unmodelled cross-excitation); regime B is correctly specified.

| quantity | Regime A (mis-specified) | Regime B (correct) |
|---|---|---|
| covariate $\delta_{\text{pop}}$ | truth 0.70, $\hat{} = 1.39$ | truth 0.60, $\hat{} = 0.675$ |
| cooldown $\delta_{\text{cool}}$ | — | truth $-0.50$, $\hat{} = -0.699$ |
| branching $\kappa$ | — | truth 0.40, $\hat{} = 0.370$ |
| decay $\theta$ | — | truth 1.00, $\hat{} = 0.819$ (weak) |
| baseline $\mu$ | — | truth 2.20, $\hat{} = 2.233$ |
| held-out IC-LL (no-cov $\to$ cov) | $18.88 \to 18.69$ | — |
| dispersion | 1.07 (inhibition tames clustering) | 2.11 (over-dispersion) |

### 7.3 A focused study: self-inhibition and lawful anti-excitation

The grid keeps self-excitation positive so the per-sector fit is well posed. We now deliberately cross the boundary of Proposition 6.3: the data-generating process has a heavy-tailed power-law kernel, *negative* self-excitation through a rectified ($\max(0, \cdot)$) intensity, positive cross-excitation between the similar sectors, and popularity-modulated excitation (`exp12_investments.py`, regime A). We fit the linear exponential MBPP, with and without the covariate. As a control (regime B) we generate from a process the augmented MBPP represents *exactly* — exponential kernel, covariate-modulated excitation, plus an exogenous *cooldown* covariate with a negative coefficient (lawful anti-excitation, Theorem 6.2) — and fit the same estimator. Table 2 summarizes; Figure 2 shows both.

The control is decisive: when reality matches the model, the augmented estimator recovers the branching ratio to within 0.03, the baseline to within 0.03, and both covariate coefficients — crucially including the *negative* cooldown effect, which is how anti-excitation enters lawfully — to within 0.10, with only the decay $\theta$ left loose, exactly as Remark 5.4 predicts. Under the triply mis-specified regime the covariate's sign and significance survive and modelling it improves out-of-sample likelihood, but its magnitude is inflated roughly twofold; we attribute this, as in the grid, to the short-memory exponential being forced to explain the power-law's persistence through the covariate. The dispersion numbers are a free diagnostic: the rectified self-inhibition

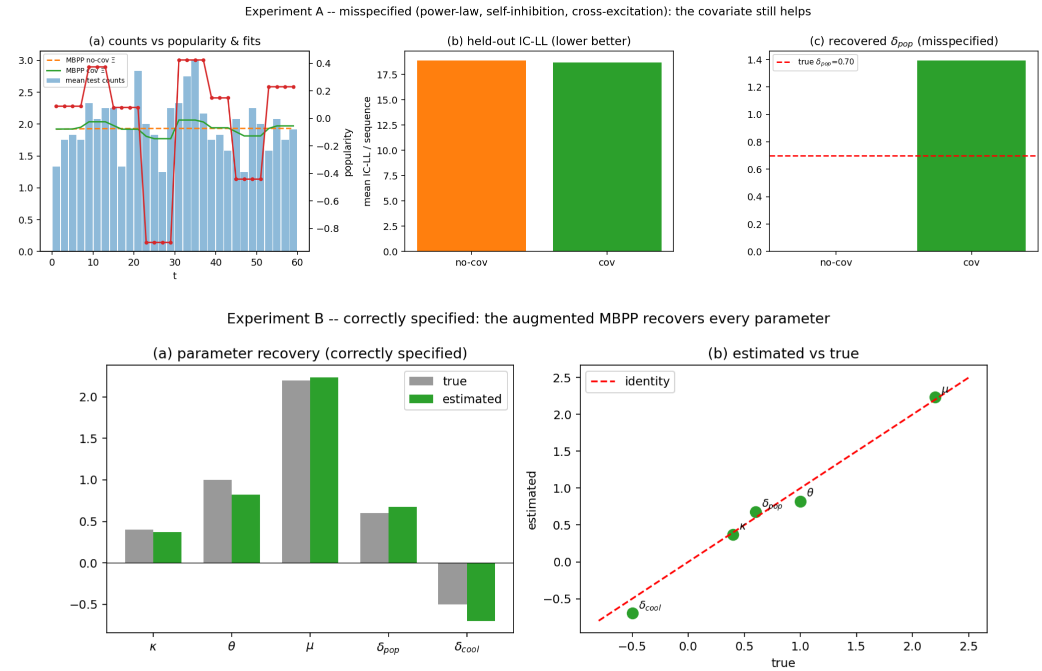


Figure 2: Top (regime A, mis-specified): counts track popularity, the covariate model improves held-out fit, and $\hat{\delta}_{\text{pop}}$ has the right sign but an inflated magnitude. Bottom (regime B, correct): every parameter — including the *negative* cooldown coefficient (lawful anti-excitation) — is recovered.

makes regime A *less* over-dispersed than the genuinely branching regime B, a sanity check that the two regimes differ in the way the model says they should.

### 7.4 Genuinely multivariate estimation

The estimator `fit_mbpp_ic_excitation_multi` fits the full construction of Theorem 6.2 in $M$ dimensions: an $M$-vector baseline, an $M \times M$ branching matrix, a shared decay, and a covariate vector $\delta$, with the compensator from the multivariate LTV solver. Two checks. First, *exactness of the reduction*: with $\delta = 0$ the covariate solver coincides with the constant-excitation multivariate ODE to $8.9 \times 10^{-16}$, confirming the forward model is correct. Second, *recovery*: from interval counts drawn from the model the estimator recovers the baseline vector and the covariate effect with the correct sign and the overall branching (spectral radius ˆ = 0.31 vs. truth 0.30). The full $M \times M$ *structure* — which sector excites which — is, however, only weakly identified from counts at moderate sample sizes: self- and cross-excitation both raise a sector's counts, and only cross-interval correlations disambiguate them. This is the multivariate face of Remark 5.4: levels and the covariate are identified, fine structure needs more data (or event-time information). The estimator is wired and validated; publication-scale recovery of the matrix is a matter of sample size and compute, not of the method.

### 7.5 Learned solvers (work in progress)

Finally, the same family is the testbed for replacing the numerical solver with a *learned* one — a Fourier neural operator or DeepONet for the forward map, a physics-informed network (PINN/PINO) trained on the MBPP residual as a fast differentiable surrogate inside the fit, and an amortized network for the inverse map counts $\mapsto (\kappa, \theta, \delta)$. The enabling fact is proved in Section 6 and re-confirmed numerically here: the MBPP residual is *linear in* $\xi$, so its global minimum is the exact solution. We verify that the residual evaluated at exact solutions is $1.4 \times 10^{-14}$ and that the residual's zero reproduces the Volterra solve to $6 \times 10^{-15}$ — i.e. the objective a PINN/PINO minimizes is well posed across the whole parameter family. The full protocol (datasets, models, metrics, success criteria) is specified in `PURE_LEARNING.md` with runnable scaffolding in `experiments/learning/`; those runs require JAX/TensorFlow and are deferred to that pipeline.

## 8 Conclusion

The Mean Behavior Poisson Process turns an intractable problem — fitting a self-exciting process to data with no event times — into a tractable one, by replacing the random Hawkes intensity with its deterministic mean and invoking Watanabe's theorem to recover a genuine Poisson likelihood. We have reproduced the foundational results in full and then established our central methodological point with the precision a skeptic requires: covariates entering log-linearly in the baseline preserve the construction *exactly*; covariates entering the excitation preserve the mean-field identity, the Poisson structure and the likelihood, trading only the convolution (and the transform machinery that rests on it) for an exactly solvable linear time-varying ODE; and the single place the mathematics genuinely breaks — rectified, sign-indefinite self-inhibition — is identified precisely, with the sign of its bias. The experiments bear this out: under correct specification the augmented estimator recovers branching ratios and covariate effects (including lawful anti-excitation) cleanly; under mis-specification it degrades in the way the identifiability analysis predicts, gracefully for timescale errors and sharply for an unrepresentable heavy tail. The honest limits — the weakly identified decay, the weakly identified multivariate fine structure, the Poisson/Hawkes degeneracy under a free baseline — are stated as plainly as the successes.

**Acknowledgement of sources.** Sections 2–5 reproduce, with proofs, the framework of Rizoiu et al. (2022), *Interval-censored Hawkes processes*, JMLR 23(1); the covariate augmentation of Section 6, the precise nonlinear-inhibition boundary, and all experiments are the present contribution.

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